

Name: Charlie Pyle

NUID: 22821006

Instructions Follow instructions *carefully*, failure to do so will result in points being deducted. Be sure to show sufficient work to justify your answer(s).

1. 12 points For each of these relations on the set {ants, bees, crickets, dragonflies}, decide whether it is reflexive, symmetric, antisymmetric, and/or transitive. **Justify** your answer.
- (a) {(bees, bees), (bees, crickets), (bees, dragonflies), (crickets, bees), (crickets, crickets), (crickets, dragonflies)}
- (b) {(ants, ants), (ants, bees), (bees, ants), (bees, bees), (crickets, crickets), (dragonflies, ants), (dragonflies, dragonflies)}
- (c) {(ants, ants), (bees, bees), (crickets, crickets), (dragonflies, dragonflies)}

Solution: Enter your solution here:

Let $A = \text{ants}$, $B = \text{bees}$, $C = \text{crickets}$, $D = \text{dragonflies}$.

A:

$R_1 = \{(B, B), (B, C), (B, D), (C, B), (C, C), (C, D)\}$

Reflexive: No, missing (A, A) and (D, D) .

Symmetric: No, we have $(B, D) \in R_1$ but $(D, B) \notin R_1$.

Antisymmetric: No, we have $(B, C) \in R_1$ and $(C, B) \in R_1$.

Transitive: Yes, checking all required pairs:

- (B, B) and (B, C) implies (B, C) ✓
- (B, B) and (B, D) implies (B, D) ✓
- (B, C) and (C, B) implies (B, B) ✓
- (B, C) and (C, C) implies (B, C) ✓
- (B, C) and (C, D) implies (B, D) ✓
- (C, B) and (B, B) implies (C, B) ✓
- (C, B) and (B, C) implies (C, C) ✓
- (C, B) and (B, D) implies (C, D) ✓
- (C, C) and (C, B) implies (C, B) ✓
- (C, C) and (C, D) implies (C, D) ✓

All required pairs are present.

B:

$R_2 = \{(A, A), (A, B), (B, A), (B, B), (C, C), (D, A), (D, D)\}$

Reflexive: Yes, contains all pairs (x, x) for every element x in the set.

Symmetric: No, we have $(D, A) \in R_2$ but $(A, D) \notin R_2$.

Antisymmetric: No, we have $(A, B) \in R_2$ and $(B, A) \in R_2$.

Transitive: No, we have $(D, A) \in R_2$ and $(A, B) \in R_2$, but $(D, B) \notin R_2$.

C:

$R_3 = \{(A, A), (B, B), (C, C), (D, D)\}$

Reflexive: Yes, contains all pairs (x, x) for every element x in the set.

Symmetric: Yes, for every $(x, y) \in R_3$, we have $x = y$, so $(y, x) = (x, x) \in R_3$.

Antisymmetric: Yes, for any $(x, y), (y, x) \in R_3$ with $x \neq y$, there are no such pairs since all pairs have the form (x, x) .

Transitive: Yes, this relation contains only pairs where an element is related to itself.

2. 12 points Determine whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

(a) $y = x$.

(b) $y = x^2$.

(c) $y < x^2$.

Justify your answer.

Solution: Enter your solution here:

A:

Reflexive: For (x, x) to be in R , we need $x = x$ which is always true.

Symmetric: Suppose $(x, y) \in R$, then $y = x$. For $(y, x) \in R$, we need $x = y$, which is true since $y = x$.

Antisymmetric: Yes, for any $(x, y), (y, x) \in R$ with $x \neq y$, there are no such pairs since all pairs have the form (x, x) .

Transitive: Yes, this relation contains only pairs where an element is related to itself.

B:

Reflexive: For (x, x) to be in R , we need $x = x^2$ which means $x^2 - x = 0$. This is only true for $x = 0$ or $x = 1$.

Counterexample: $(2, 2) \notin R$ because $2 \neq 2^2$ since $2^2 = 4$, therefore R is not reflexive.

Symmetric: Suppose $(x, y) \in R$ and $(y, x) \in R$. From $(x, y) \in R$: $y = x^2$. From $(y, x) \in R$: $x = y^2 = (x^2)^2 = x^4$. This is only true when $x^4 = x^2$, i.e., $x \in \{0, 1, -1\}$.

Counterexample: $(2, 4) \in R$ (since $4 = 2^2$), but $(4, 2) \notin R$ (since $2 \neq 4^2 = 16$), therefore R is not symmetric.

Antisymmetric: Suppose $(x, y) \in R$ and $(y, x) \in R$. From $(x, y) \in R$: $y = x^2$. From $(y, x) \in R$: $x = y^2 = (x^2)^2 = x^4$. From $x = x^4$, we get $x^4 - x = 0$, so $x(x^3 - 1) = 0$, giving $x \in \{0, 1, -1\}$. If $x = 0$: then $y = 0^2 = 0$, so $x = y$. If $x = 1$: then $y = 1^2 = 1$, so $x = y$. If $x = -1$: then $y = (-1)^2 = 1$, but for $(1, -1) \in R$ we need $-1 = 1^2 = 1$, which is false. So $(1, -1) \notin R$, therefore R is antisymmetric.

Transitive: Suppose $(x, y) \in R$ and $(y, z) \in R$. From $(x, y) \in R$: $y = x^2$. From $(y, z) \in R$: $z = y^2 = (x^2)^2 = x^4$. For $(x, z) \in R$, we need $z = x^2$, but we have $z = x^4$. This is only true when $x^4 = x^2$, i.e., $x \in \{0, 1, -1\}$.

Counterexample: $(2, 4) \in R$ and $(4, 16) \in R$, but $(2, 16) \notin R$ since $16 \neq 2^2 = 4$, therefore R is not transitive.

C:

Reflexive: For $(x, x) \in R$, we need $x < x^2$, which means $x^2 - x > 0$, so $x(x - 1) > 0$. This is true when $x < 0$ or $x > 1$.

Counterexample: $(0, 0) \notin R$ because $0 \not< 0^2 = 0$, therefore R is not reflexive.

Symmetric: Suppose $(x, y) \in R$, so $y < x^2$. For $(y, x) \in R$, we need $x < y^2$.

Counterexample: $(2, 1) \in R$ since $1 < 2^2 = 4$, but $(1, 2) \notin R$ since $2 \not< 1^2 = 1$, therefore R is not symmetric.

Antisymmetric: Suppose $(x, y) \in R$ and $(y, x) \in R$, so $y < x^2$ and $x < y^2$. From $x < y^2$, we get $x^2 < y^4$. Combined with $y < x^2$, we have $y < x^2 < y^4$. This gives $y < y^4$, so $y^4 - y > 0$, meaning $y(y - 1)(y^2 + y + 1) > 0$. Similarly, from $y < x^2$, we get $y^2 < x^4$, and with $x < y^2$, we have $x < y^2 < x^4$. If $x \neq y$, these conditions cannot be satisfied simultaneously, therefore R is antisymmetric.

Transitive: Suppose $(x, y) \in R$ and $(y, z) \in R$. Then $y < x^2$ and $z < y^2$. For $(x, z) \in R$, we need $z < x^2$. We have $z < y^2 < (x^2)^2 = x^4$, so $z < x^4$, but this doesn't guarantee $z < x^2$.

Counterexample: $(2, 3) \in R$ (since $3 < 4$) and $(3, 8) \in R$ (since $8 < 9$), but $(2, 8) \notin R$ since $8 \not< 2^2 = 4$, therefore R is not transitive.

3. 6 points Represent each of these relations on $\{A, B, C, D\}$ with a matrix (with the elements of this set listed in increasing order).

(a) $\{(A, A), (A, B), (B, C), (B, D), (C, C), (C, D), (D, D), (D, A)\}$

(b) $\{(A, A), (A, D), (B, B), (B, C), (C, B), (C, C), (D, A), (D, D)\}$

Solution: Enter your solution here:

A:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

B:

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

4. 6 points Consider the set $\{1, 2, 3, 4\}$. Find the transitive closure on this set of the relation

$$\{(1, 1), (1, 4), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 1), (4, 3)\}$$

using Warshall's algorithm. **You must show the steps of your calculation.**

Solution: Enter your solution here:

Initial Matrix:

$$W_0 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Step $k = 1$:

$$W_1 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Step $k = 2$:

$$W_2 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Step $k = 3$:

$$W_3 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Step $k = 4$:

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Final Transitive Closure:

$$W^* = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

5. 6 points Which of these relations on the set of all people are equivalence relations? Determine the properties of an equivalence relation that the others lack. **Justify** your answer.
- (a) $\{(a, b) \mid a \text{ and } b \text{ have eaten breakfast together}\}$
 - (b) $\{(a, b) \mid a \text{ and } b \text{ share a common last name}\}$

Solution: Enter your solution here:

A:

The relation "a and b have eaten breakfast together" is:

Reflexive - Yes: Everyone has eaten breakfast with themselves

Symmetric - Yes: If a ate with b, then b ate with a

Transitive - No: a eating with b and b eating with c, does not imply a ate with c

Relation (a) is not an equivalence relation since it is not transitive.

B:

The relation "a and b share a common last name" is:

Reflexive - Yes: everyone shares their last name with themselves

Symmetric - Yes: if a shares a last name with b, then b shares a last name with a

Transitive - Yes: sharing the same last name is transitive

Relation (b) is an equivalence relation since it is reflexive, symmetric, and transitive.

6. 10 points Given the relation on the set $\{A, B, C, D\}$, which is represented by the following zero-one matrix

(a) draw the corresponding directed graph.

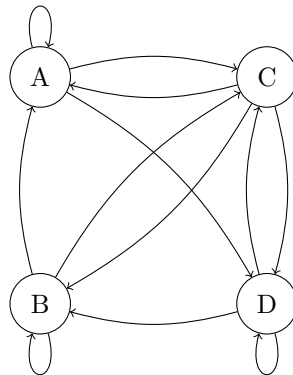
(b) determine whether it is an equivalence relation. **Justify** your answer.

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Solution: Enter your solution here:

A:

Directed graph for the relation:



B:

NO, this is NOT an equivalence relation.

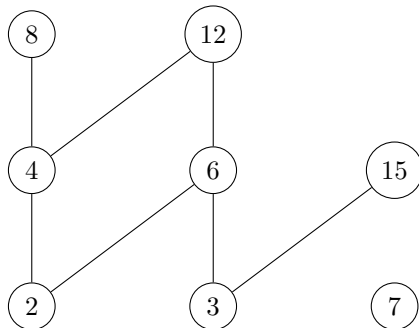
Reflexive: No. The relation is missing (C, C) . Looking at the diagonal of the matrix, position $[3,3] = 0$, so C is not related to itself.

Symmetric: No. We have $(A, D) \in R$ (matrix entry $[1,4] = 1$) but $(D, A) \notin R$ (matrix entry $[4,1] = 0$), thus the relation fails symmetry due to the (A, D) counterexample.

Since the relation fails both reflexivity and symmetry, it cannot be an equivalence relation.

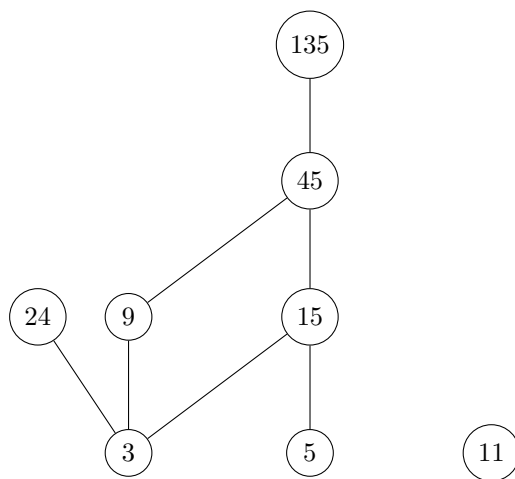
7. 4 points Draw the Hasse diagram for divisibility on the set $\{2, 3, 4, 6, 7, 8, 12, 15\}$.

Solution: Enter your solution here:



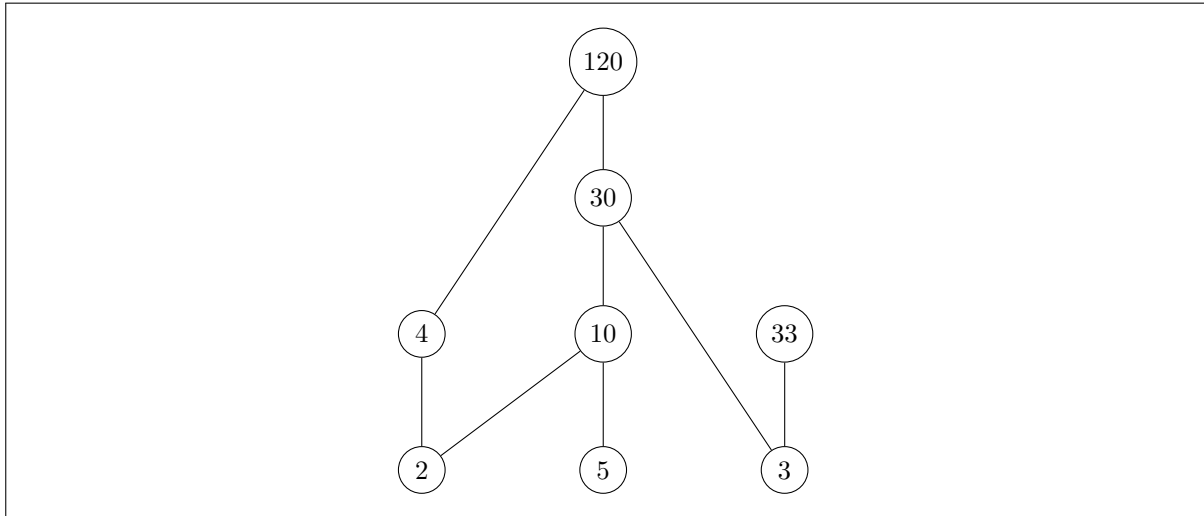
8. 4 points Draw the Hasse diagram for divisibility on the set $\{3, 5, 9, 11, 15, 24, 45, 135\}$.

Solution: Enter your solution here:



9. 4 points Draw the Hasse diagram for divisibility on the set $(\{2, 3, 4, 5, 10, 30, 33, 120\}, |)$.

Solution: Enter your solution here:



10. 16 points Answer these questions for the poset $\{3, 5, 9, 11, 15, 24, 45, 135\}$. **Hint:** Use your Hasse diagram from question 8.
- Find the maximal elements.
 - Find the minimal elements.
 - Is there a greatest element?
 - Is there a least element?
 - Find all upper bounds of $\{3, 5\}$.
 - Find the least upper bound of $\{3, 5\}$, if it exists.
 - Find all lower bounds of $\{15, 135\}$.
 - Find the greatest lower bound of $\{15, 135\}$, if it exists.

Solution: Enter your solution here:

A: 11, 24, 135

B: 3, 5, 11

C: There is not since there is multiple maximal elements.

D: There is not since there is multiple minimal elements.

E: 15, 45, 135

F: 15

G: 3, 5, 15

H: 15

11. 16 points Answer these questions for the poset $(\{2, 3, 4, 5, 10, 30, 33, 120\}, |)$. **Hint:** Use your Hasse diagram from question 9.
- Find the maximal elements.
 - Find the minimal elements.
 - Is there a greatest element?
 - Is there a least element?

- (e) Find all upper bounds of $\{2, 5\}$.
- (f) Find the least upper bound of $\{2, 5\}$, if it exists.
- (g) Find all lower bounds of $\{30, 120\}$.
- (h) Find the greatest lower bound of $\{30, 120\}$, if it exists.

Solution: Enter your solution here:

A: 33, 120

B: 2, 3, 5

C: There is not since there is multiple maximal elements.

D: There is not since there is multiple minimal elements.

E: 10, 30, 120

F: 10

G: 2, 3, 5, 10, 30

H: 30

12. 4 points Find a compatible total order using topological sorting for the divisibility relation on the set $\{1, 2, 3, 6, 10, 12, 20, 32, 40, 49\}$.

Solution: Enter your solution here:

A possible total order from topological sorting is (1, 49, 2, 3, 32, 10, 6, 20, 40, 12).